

Blueway1721 Datasheet Rx beamforming IC

General description

Sivers Semiconductors' Blueway1721 is a highly integrated multi beam-forming front-end receive IC, targeting phased-array receive antennas for satellite communication. The device supports two dual-polarization elements and two reconfigurable and independent beam directions. The device covers the 17.7 to 21.2 GHz Satcom receive band and offers exceptionally low noise figure, low power, and high integration with on-chip beam table memory. The low NF and low-power reconfigurable multi-beamforming architectures enables Blueway1721-based arrays to operate with smaller size, lower cost and support higher data rates.

Applications

- Satellite communications

Key specifications

- Four-element dual-polarization (8 Total Channels)
- 45RFSOI CMOS technology
- Reconfigurable and independent two beam directions
- Ultra-low leakage in standby mode; Low power consumption in active mode
- Phase Shifting Control
 - 6-bit full 360° range per path
 - 5.625° step
- Gain Control
 - 6-bit 32-dB range per path
 - 0.5-dB step
- 16-dB common-beam gain control with 0.25-dB step
- On-chip programable PTAT biasing
- On-chip temperature sensor
- On-chip power sensor for two beam outputs
- 1V, 1.5V, and 1.8V power supplies
- Up to 100MHz SPI control
- 1024-entry on-chip beam table storage
- 8-bit chip addressing supports 1024 element dual polarization arrays
- Chip-Scale Package (CSP)
 - 5.2 mm x 5.2 mm
- ESD protection on all pins

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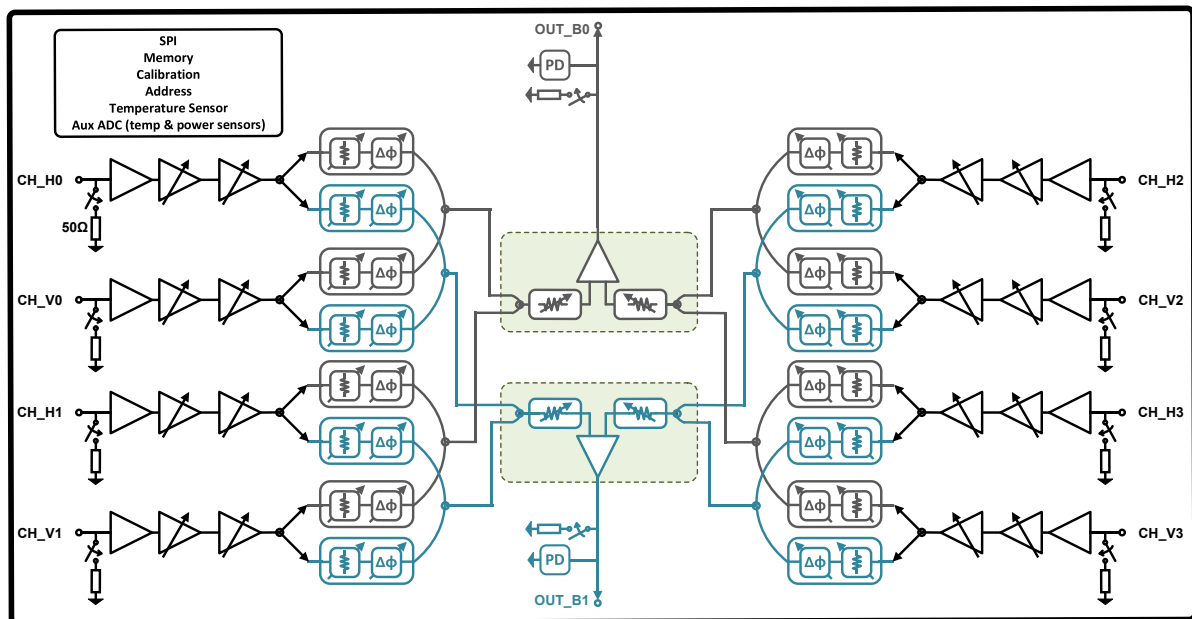
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1 Overview

Sivers Semiconductors' Blueway1721 is a highly integrated multi beam-forming front-end receive IC, targeting phased-array receive antennas for satellite communication. The device supports two dual-polarization elements and two reconfigurable and independent beam directions. The device covers the 17.7 to 21.2 GHz Satcom receive band and offers exceptionally low noise figure, low power, and high integration with on-chip beam table memory. The low NF and low-power reconfigurable multi-beamforming architectures enable Blueway1721-based arrays to operate with smaller size, lower cost, and support higher data rates.

Figure 2.1: Blueway1721 Block Diagram



2 Pin Configuration

2.1 Pin Diagram

Figure 2.1.1: Bump Pad Diagram with bump down view shows the bump pad locations for 92-bump WLCSP package.

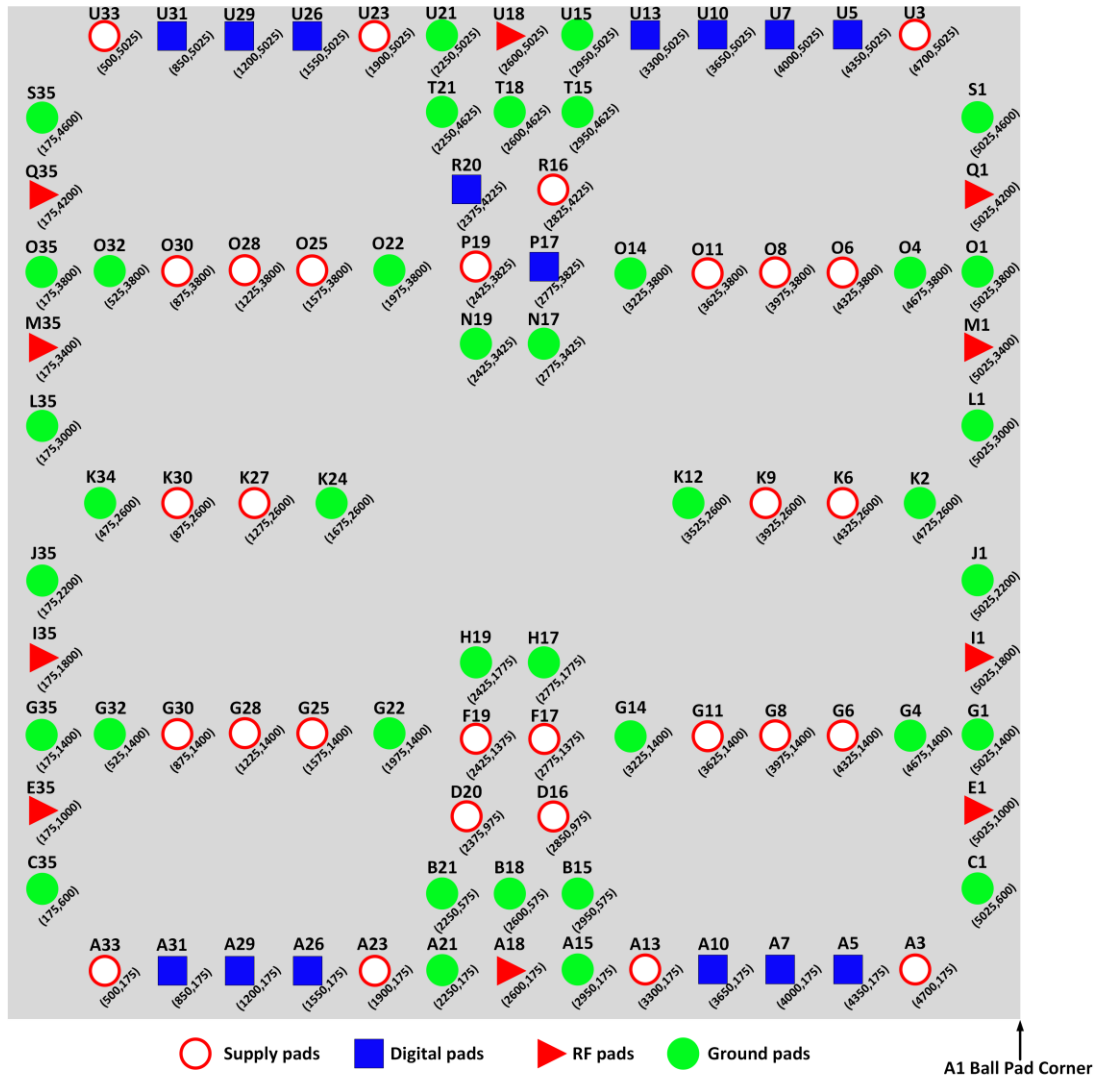
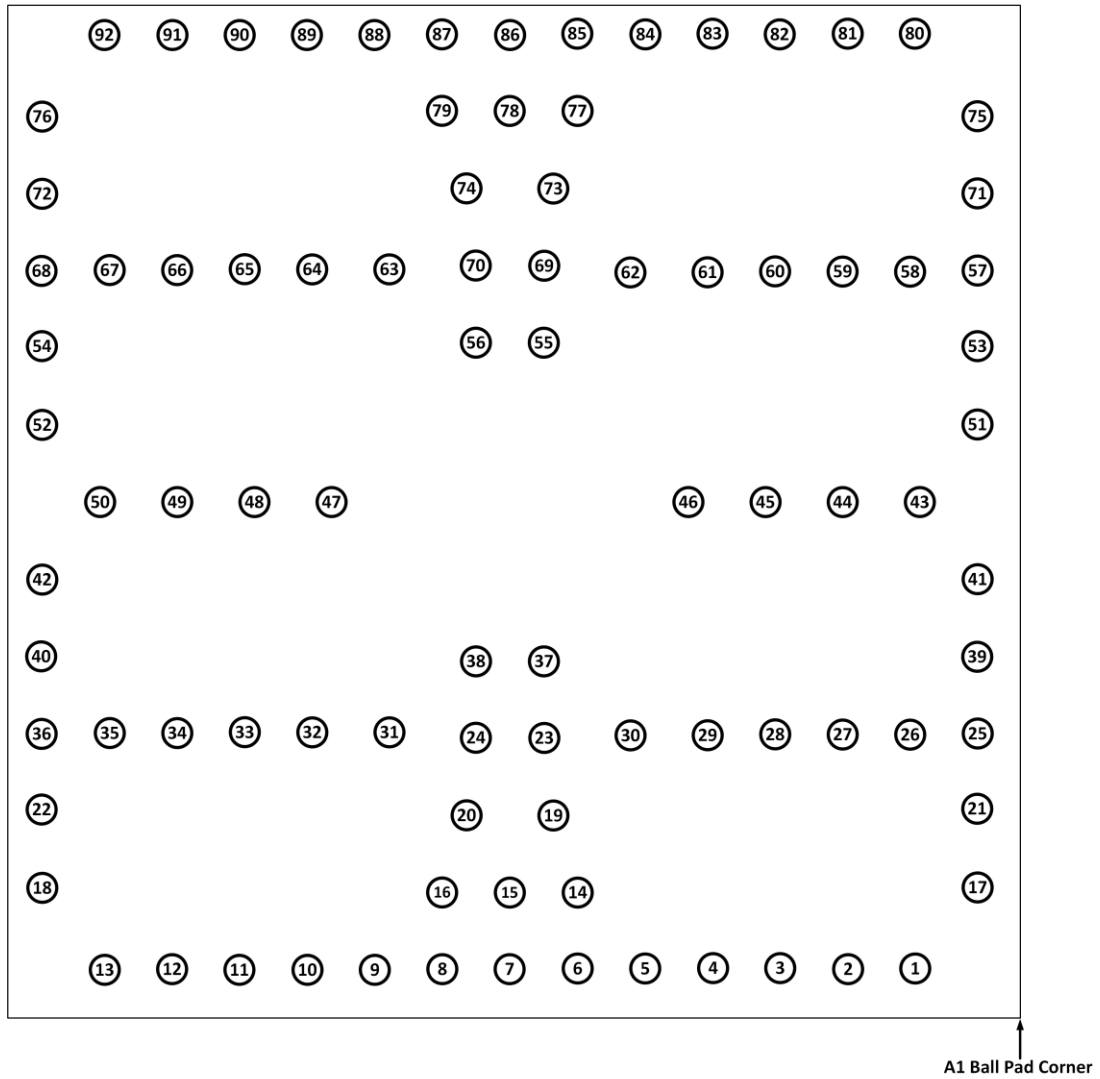


Figure 2.1.2: The pad ID assignments for 92-pin WLCSP package with bump-down view.



2.2 Pin List, Description, and Location

Pad ID	BGA ID	Pad Center (μm)		Pad Name	IO Type	Description
		X	Y			
1	A3	4700	175	VDD_FE3_CH1	POWER SUPPLY INPUT	1.5-V POWER SUPPLY FOR CH1 FE3 SPLITTER
2	A5	4350	175	SPI_CHIPID<0>	DIGITAL INPUT	DIGITAL SPI CHIP ID<0>
3	A7	4000	175	SPI_CHIPID<1>	DIGITAL INPUT	DIGITAL SPI CHIP ID<1>
4	A10	3650	175	SPI_CHIPID<2>	DIGITAL INPUT	DIGITAL SPI CHIP ID<2>
5	A13	3300	175	VDD_1p8V_ANA	POWER SUPPLY INPUT	1.8-V ANALOG POWER SUPPLY
6	A15	2950	175	GND	GROUND	GROUND
7	A18	2600	175	RF_Beam1	RF OUTPUT	RF OUTPUT BEAM1
8	A21	2250	175	GND	GROUND	GROUND
9	A23	1900	175	DCTEST_VOUT	ANALOG OUTPUT	DCTEST ANALOG OUTPUT
10	A26	1550	175	SPI_RESET_N	DIGITAL INPUT	DIGITAL SPI RESET_N WITHOUT ANY PULLUP/PULLDOWN ENABLE
11	A29	1200	175	CHAIN_CLK	DIGITAL INPUT	DIGITAL SPI CHAIN CLK
12	A31	850	175	CHAIN_MOSI	DIGITAL INPUT	DIGITAL SPI CHAIN MOSI
13	A33	500	175	VDD_FE3_CH3	POWER SUPPLY INPUT	1.5-V POWER SUPPLY FOR CH3 FE3 SPLITTER
14	B15	2950	575	GND	GROUND	GROUND
15	B18	2600	575	GND	GROUND	GROUND
16	B21	2250	575	GND	GROUND	GROUND
17	C1	5025	600	GND	GROUND	GROUND
18	C35	175	600	GND	GROUND	GROUND
19	D16	2825	975	EFUSE_FSOURCE	POWER SUPPLY INPUT	DIGITAL SPI 1.5-V EFUSE FSOURCE
20	D20	2375	975	VDD_1p8V_ANA	POWER SUPPLY INPUT	1.8-V ANALOG POWER SUPPLY
21	E1	5025	1000	RF_CH1_VPOL	RF INPUT	SINGLE-ENDED RF INPUT CH1 VPOL
22	E35	175	1000	RF_CH3_VPOL	RF INPUT	SINGLE-ENDED RF INPUT CH3 VPOL
23	F17	2775	1375	VDD_1p8V_IO_SOUTH	POWER SUPPLY INPUT	1.8-V DIGITAL IO POWER SUPPLY AT SOUTH
24	F19	2425	1375	VDD_DIG	POWER SUPPLY INPUT	1.0-V DIGITAL POWER SUPPLY

25	G1	5025	1400	GND	GROUND	GROUND
26	G4	4675	1400	GND	GROUND	GROUND
27	G6	4325	1400	VDD_FE1_CH1	POWER SUPPLY INPUT	1-V POWER SUPPLY FOR CH1 FE1 LNA ST1
28	G8	3975	1400	VDD_FE2_CH1	POWER SUPPLY INPUT	1-V POWER SUPPLY FOR CH1 FE2 LNA ST2
29	G11	3625	1400	VDD_DIG	POWER SUPPLY INPUT	1.0-V DIGITAL POWER SUPPLY
30	G14	3225	1400	GND	GROUND	GROUND
31	G22	1975	1400	GND	GROUND	GROUND
32	G25	1575	1400	VDD_DIG	POWER SUPPLY INPUT	1.0-V DIGITAL POWER SUPPLY
33	G28	1225	1400	VDD_FE2_CH3	POWER SUPPLY INPUT	1-V POWER SUPPLY FOR CH3 FE2 LNA ST2
34	G30	875	1400	VDD_FE1_CH3	POWER SUPPLY INPUT	1-V POWER SUPPLY FOR CH3 FE1 LNA ST1
35	G32	525	1400	GND	GROUND	GROUND
36	G35	175	1400	GND	GROUND	GROUND
37	H17	2775	1775	GND	GROUND	GROUND
38	H19	2425	1775	GND	GROUND	GROUND
39	I1	5025	1800	RF_CH1_HPOL	RF INPUT	SINGLE-ENDED RF INPUT CH1 HPOL
40	I35	175	1800	RF_CH3_HPOL	RF INPUT	SINGLE-ENDED RF INPUT CH3 HPOL
41	J1	5025	2200	GND	GROUND	GROUND
42	J35	175	2200	GND	GROUND	GROUND
43	K2	4725	2600	GND	GROUND	GROUND
44	K6	4325	2600	VDD_1p8V_ANA	POWER SUPPLY INPUT	1.8-V ANALOG POWER SUPPLY
45	K9	3925	2600	VDD_1p8V_DIST	POWER SUPPLY INPUT	1.5-V POWER SUPPLY FOR RF DIST COMBINER
46	K12	3525	2600	GND	GROUND	GROUND
47	K24	1675	2600	GND	GROUND	GROUND
48	K27	1275	2600	VDD_1p8V_DIST	POWER SUPPLY INPUT	1.5-V POWER SUPPLY FOR RF DIST COMBINER
49	K30	875	2600	VDD_1p8V_ANA	POWER SUPPLY INPUT	1.8-V ANALOG POWER SUPPLY
50	K34	475	2600	GND	GROUND	GROUND
51	L1	5025	3000	GND	GROUND	GROUND

52	L35	175	3000	GND	GROUND	GROUND
53	M1	5025	3400	RF_CH0_VPOL	RF INPUT	SINGLE-ENDED RF INPUT CH0 VPOL
54	M35	175	3400	RF_CH2_VPOL	RF INPUT	SINGLE-ENDED RF INPUT CH2 VPOL
55	N17	2775	3425	GND	GROUND	GROUND
56	N19	2425	3425	GND	GROUND	GROUND
57	O1	5025	3800	GND	GROUND	GROUND
58	O4	4675	3800	GND	GROUND	GROUND
59	O6	4325	3800	VDD_FE1_CH0	POWER SUPPLY INPUT	1-V POWER SUPPLY FOR CH1 FE0 LNA ST1
60	O8	3975	3800	VDD_FE2_CH0	POWER SUPPLY INPUT	1-V POWER SUPPLY FOR CH0 FE2 LNA ST2
61	O11	3625	3800	VDD_DIG	POWER SUPPLY INPUT	1.0-V DIGITAL POWER SUPPLY
62	O14	3225	3800	GND	GROUND	GROUND
63	O22	1975	3800	GND	GROUND	GROUND
64	O25	1575	3800	VDD_DIG	POWER SUPPLY INPUT	1.0-V DIGITAL POWER SUPPLY
65	O28	1225	3800	VDD_FE2_CH2	POWER SUPPLY INPUT	1-V POWER SUPPLY FOR CH2 FE2 LNA ST2
66	O30	875	3800	VDD_FE1_CH2	POWER SUPPLY INPUT	1-V POWER SUPPLY FOR CH1 FE2 LNA ST1
67	O32	525	3800	GND	GROUND	GROUND
68	O35	175	3800	GND	GROUND	GROUND
69	P17	2775	3825	SPI_MISO	DIGITAL INPUT	DIGITAL SPI MISO
70	P19	2425	3825	VDD_DIG	POWER SUPPLY INPUT	1.0-V DIGITAL POWER SUPPLY
71	Q1	5025	4200	RF_CH0_HPOL	RF INPUT	SINGLE-ENDED RF INPUT CH0 HPOL
72	Q35	175	4200	RF_CH2_HPOL	RF INPUT	SINGLE-ENDED RF INPUT CH2 HPOL
73	R16	2825	4225	VDD_1p8V_IO_NORTH	POWER SUPPLY INPUT	1.8-V DIGITAL IO POWER SUPPLY AT NORTH
74	R20	2375	4225	Beam_UPDATE	DIGITAL INPUT	DIGITAL SPI BEAM UPDATE
75	S1	5025	4600	GND	GROUND	GROUND
76	S35	175	4600	GND	GROUND	GROUND
77	T15	2950	4625	GND	GROUND	GROUND
78	T18	2600	4625	GND	GROUND	GROUND

79	T21	2250	4625	GND	GROUND	GROUND
80	U3	4700	5025	VDD_FE3_CH0	POWER SUPPLY INPUT	1.5-V POWER SUPPLY FOR CH0 FE3 SPLITTER
81	U5	4350	5025	CHAIN_CSN	DIGITAL INPUT	DIGITAL SPI CHAIN CSN
82	U7	4000	5025	SPI_CSN	DIGITAL INPUT	DIGITAL SPI CSN
83	U10	3650	5025	SPI_CLK	DIGITAL INPUT	DIGITAL SPI CLK
84	U13	3300	5025	SPI_MOSI	DIGITAL INPUT	DIGITAL SPI MOSI
85	U15	2950	5025	GND	GROUND	GROUND
86	U18	2600	5025	RF_Beam0	RF OUTPUT	RF OUTPUT BEAM0
87	U21	2250	5025	GND	GROUND	GROUND
88	U23	1900	5025	VDD_1p8V_ANA	POWER SUPPLY INPUT	1.8-V ANALOG POWER SUPPLY
89	U26	1550	5025	SPI_CHIPID<3>	DIGITAL INPUT	DIGITAL SPI CHIP ID<3>
90	U29	1200	5025	SPI_CHIPID<4>	DIGITAL INPUT	DIGITAL SPI CHIP ID<4>
91	U31	850	5025	SPI_CHIPID<5>	DIGITAL INPUT	DIGITAL SPI CHIP ID<5>
92	U33	500	5025	VDD_FE3_CH2	POWER SUPPLY INPUT	1.5-V POWER SUPPLY FOR CH2 FE3 SPLITTER

3 Electrical Characteristics

3.1 Power Supply Voltage and Current Limits

Stressing the device beyond the conditions shown in Table 2 and Table 3 may cause permanent damage to the device.

Table 1: Power Supply Rail Characteristics

Supply	Device blocks powered from the supply	Relevant IOs in the device
1.0V	1 st -stage LNA	VDD_FE1<3:0>
1.0V	2 nd -stage LNA	VDD_FE2<3:0>
1.5V	3 rd -stage splitter and active combiners	VDD_FE3 and VDD_1p5V_DIST
1.8V	Level shifters, Temp sensor, ADCs, LDOs, Bandgap, Power Detectors	VDD_1p8V_ANA
1.0V	Digital SPI	VDD_DIG
1.8V	Digital IOs	VDD_1p8V_IO

Table 2: Power supply rail ratings operation level

Supply Name	Descriptions	Min	Nom	Max	Unit
VDD_FE1<3:0>	1.0-V RF power supply for the LNA St1 and St2	0.95	1	1.05	V
VDD_FE3<1:0>	1.5-V RF power supply for the splitter	1.4	1.5	1.6	V
VDD_1p5V_DIST	1.5-V RF power supply for the combiners St1 and St2	1.4	1.5	1.6	V
VDD_1p8V_ANA	1.8-V analog power supply	1.7	1.8	1.9	V
VDD_DIG	1.0-V digital power supply	0.95	1	1.05	V
VDD_1p8V_IO	1.8-V digital power supply	1.7	1.8	1.9	V

Table 3: Power supply rail ratings, absolute maximum ratings

Parameter	Descriptions	Max	Units
VDD_FE1<3:0>	1.0-V RF power supply for the LNA St1 and St2	1.1	V
VDD_FE3<1:0>	1.5-V RF power supply for the splitter	1.8	V
VDD_1p5V_DIST	1.5-V RF power supply for the combiners St1 and St2	1.8	V
VDD_1p8V_ANA	1.8-V analog power supply	2.1	V
VDD_DIG	1.0-V digital power supply	1.5	V
VDD_1p8V_IO	1.8-V digital power supply	2.1	V

Table 4: Maximum Current Ratings at Power Terminals

Supply Name	Descriptions	Min*	Nom*	Max*	Unit
VDD_FE1<3:0>, VDD_DIG	Total current drawn by all nodes driven by 1.0V rail	200	300	450	mA
VDD_FE3<1:0>, VDD_1p5V_DIST	Total current drawn by all nodes driven by 1.5V rail	80	130	250	mA
VDD_1p8V_ANA, VDD_1p8V_IO	Total current drawn by all nodes driven by 1.8V rail	10	20	50	mA

*Simulated data; min are results at SS (Sigma of -2), -40C, and at -5% supply; nom are results at TT (Sigma of 0), 27C, and nominal supply; max are results at FF (Sigma of +2), 125C, and +5% supply.

3.2 Absolute Maximum for RF Pads

Table 5: Abs Max for RF Pads

Symbol	Parameter	Min	Typ	Max	Unit
P_{RF-ABS}	RF input power at any RF pin			10	dBm
T_J	Junction Temperature			120	°C
T_{ST}	Storage Temperature Range	-40		150	°C
T_{LEAD}	Lead temperature (soldering, 10s)		220		°C

3.3 ESD Pads

Table 6: ESD Ratings

Symbol	Parameter	Min	Typ	Max	Unit
ESD HBM	Electrostatic Discharge – Human Body Model		1000		V
ESD HBM RF	Electrostatic Discharge – Human Body Model – RF pins		500		V
ESD CDM	Electrostatic Discharge – Charged Device Model		125		V

4 RF Specifications

This section describes the RF characteristics of the BFC.

4.1 Receiver Specifications

Table 7: Receiver Specifications

Parameter	Conditions	Min	Nom	Max	Units
Frequency range		17.7		21.2	GHz
No. of Beams			2		
No. of Channels			8		
Gain	Single-Channel, one beam	15	18	21	dB
Gain	All channel one beam/dual beam	33	36	39	dB
NF	Single-Channel, one beam		2	2.3	dB
NF	All channel one beam/dual beam		2	2.3	dB
IP1dB	Single-Channel _At max gain setting. The iP1dB can be improved by engaging attenuation	-36	-33	-30	dBm
IP1dB	All channels activated-2beam _At max gain setting. The iP1dB can be improved by engaging attenuation	TBD	TBD	TBD	dBm
IIP3	Single-Channel _At max gain setting. The iP3dB can be improved by engaging attenuation	-27	-24	-21	dBm
IIP3	All channels activated-2beam _At max gain setting. The iP3dB can be improved by engaging attenuation	TBD	TBD	TBD	dBm
Phase Shift Resolution			6		Bits
Attenuator Resolution			6		Bits
Attenuator Range			32		dB
Common Attenuator Resolution			5		Bits
Common Attenuator Range			16		dB
RMS phase error		2.4	2.5	3.4	deg
RMS amplitude error		0.4	0.5	0.8	dB
Power Consumption	8 Channels and 2 beams are activated	TBD	0.7	TBD	W
Power Consumption	8 Channels, 1 beam is activated	TBD	0.6	TBD	W
Power Consumption	Single channel is activated	TBD	0.075	TBD	W
Power Consumption	Standby		0.05		W
Power Consumption	Reset		0.03		W
Second Harmonic level			TBD		dBc
Spur level			TBD		dBc
Input Return loss			-10		dB
Input Return loss			-10		dB
Isolation	Input to input		-30		dB
Isolation	Input to output		-30		dB
Isolation	Output to Output		-30		dB
Isolation	Output to input		-30		dB
Area	5.2 mm x 5.2 mm		27.0		mm ²

Table 8: Transmitter specifications thermal characteristics table

Parameter	Conditions	Min	Nom	Max	Units
θ_{JA} (Junction-to-Ambient)	Standard PCB, no airflow		TBD		°C/W
θ_{JB} (Junction-to-Board)	Measured on test board		TBD		°C/W
θ_{JC} (Junction-to-Case)	Top of package		TBD		°C/W
Total Power Dissipation	Ta = 25°C, standard PCB		TBD		W

5 Performance Summary

Table 9: Rx Control Setting in Different Modes

Bit Name	Pwr Down Mode	Thru Elem Mode	Thru Array Mode	Full Array Mode	Descriptions
		CH0HV- B0	B0	B0-1	
d2a_pwrdrn_b<5:0>	0	63	63	63	<5:3> for CH0V and <2:0> for CH0H
d2a_pwrdrn_b<11:6>	0	0	63	63	<11:9> for CH1V and <8:6> for CH1H
d2a_pwrdrn_b<17:12>	0	0	63	63	<17:15> for CH2V and <14:12> for CH2H
d2a_pwrdrn_b<23:18>	0	0	63	63	<23:21> for CH3V and <20:18> for CH3H
d2a_pwrdrn_b<32:24>	0	65	73	511	<32:30> for St1 Comb, <29:27> for CH3-2 St2_1 Comb, and <26:24> for CH1-0 St2_0 Comb
d2a_pwrdrn_b<38:33>	0	0	0	0	Reserved bits
d2a_quad_bias_en<3:0>	0	1	15	15	Quad bias enable
d2a_center_bias_en<4:0>	0	7	31	31	Center bias enable
d2a_CH0_H_multB_fe_en<2:0>	0	1	1	7	Multi-beam enable
d2a_CH0_V_multB_fe_en<2:0>	0	1	1	7	Multi-beam enable
d2a_CH1_H_multB_fe_en<2:0>	0	0	1	7	Multi-beam enable
d2a_CH1_V_multB_fe_en<2:0>	0	0	1	7	Multi-beam enable
d2a_CH2_H_multB_fe_en<2:0>	0	0	1	7	Multi-beam enable
d2a_CH2_V_multB_fe_en<2:0>	0	0	1	7	Multi-beam enable
d2a_CH3_H_multB_fe_en<2:0>	0	0	1	7	Multi-beam enable
d2a_CH3_V_multB_fe_en<2:0>	0	0	1	7	Multi-beam enable
d2a_DIST_B0_St1_en<3:0>	0	1	3	3	DIST-channel enable
d2a_DIST_B0_St2_0_en<1:0>	0	1	3	3	DIST-channel enable
d2a_DIST_B0_St2_1_en<1:0>	0	0	3	3	DIST-channel enable
d2a_DIST_B1_St1_en<3:0>	0	0	0	3	DIST-channel enable
d2a_DIST_B1_St2_0_en<1:0>	0	0	0	3	DIST-channel enable
d2a_DIST_B1_St2_1_en<1:0>	0	0	0	3	DIST-channel enable
d2a_match_ctrl_B0	1	0	0	0	Output term enable
d2a_match_ctrl_B1	1	1	1	0	Output term enable
d2a_CH0_H_PTAT_St1<5:0>	0	24	24	24	PTAT bias control
d2a_CH0_H_PTAT_St2<5:0>	0	24	24	24	PTAT bias control
d2a_CH0_H_PTAT_St3<5:0>	0	40	24	24	PTAT bias control

Bit Name	Pwr Down Mode	Thru Elem Mode	Thru Array Mode	Full Array Mode	Descriptions
		CH0HV- B0	B0	B0-1	
d2a_CH0_V_PTAT_St1<5:0>	0	24	24	24	PTAT bias control
d2a_CH0_V_PTAT_St2<5:0>	0	24	24	24	PTAT bias control
d2a_CH0_V_PTAT_St3<5:0>	0	40	24	24	PTAT bias control
d2a_CH1_H_PTAT_St1<5:0>	0	24	24	24	PTAT bias control
d2a_CH1_H_PTAT_St2<5:0>	0	24	24	24	PTAT bias control
d2a_CH1_H_PTAT_St3<5:0>	0	40	24	24	PTAT bias control
d2a_CH1_V_PTAT_St1<5:0>	0	24	24	24	PTAT bias control
d2a_CH1_V_PTAT_St2<5:0>	0	24	24	24	PTAT bias control
d2a_CH1_V_PTAT_St3<5:0>	0	40	24	24	PTAT bias control
d2a_CH2_H_PTAT_St1<5:0>	0	24	24	24	PTAT bias control
d2a_CH2_H_PTAT_St2<5:0>	0	24	24	24	PTAT bias control
d2a_CH2_H_PTAT_St3<5:0>	0	40	24	24	PTAT bias control
d2a_CH2_V_PTAT_St1<5:0>	0	24	24	24	PTAT bias control
d2a_CH2_V_PTAT_St2<5:0>	0	24	24	24	PTAT bias control
d2a_CH2_V_PTAT_St3<5:0>	0	40	24	24	PTAT bias control
d2a_CH3_H_PTAT_St1<5:0>	0	24	24	24	PTAT bias control
d2a_CH3_H_PTAT_St2<5:0>	0	24	24	24	PTAT bias control
d2a_CH3_H_PTAT_St3<5:0>	0	40	24	24	PTAT bias control
d2a_CH3_V_PTAT_St1<5:0>	0	24	24	24	PTAT bias control
d2a_CH3_V_PTAT_St2<5:0>	0	24	24	24	PTAT bias control
d2a_CH3_V_PTAT_St3<5:0>	0	40	24	24	PTAT bias control
d2a_DIST_B0_St1_PTAT<5:0>	0	20	20	20	PTAT bias control
d2a_DIST_B0_St2_0_PTAT<5:0>	0	20	20	20	PTAT bias control
d2a_DIST_B0_St2_1_PTAT<5:0>	0	20	20	20	PTAT bias control
d2a_DIST_B1_St1_PTAT<5:0>	0	20	20	20	PTAT bias control
d2a_DIST_B1_St2_0_PTAT<5:0>	0	20	20	20	PTAT bias control
d2a_DIST_B1_St2_1_PTAT<5:0>	0	20	20	20	PTAT bias control
d2a_B0_attn<5:0>	63	30	30	0	Comm beam ATTN
d2a_B1_attn<5:0>	63	63	63	0	Comm beam ATTN

Table 10: Rx Bias Settings

Bit Name	Default codes	Descriptions
d2a_CH<X>_<H/V>_PTAT_St1	24	LNA St1 PTAT bias control
d2a_CH<X>_<H/V>_PTAT_St2	24	LNA St2 PTAT bias control
d2a_CH<X>_<H/V>_PTAT_St3	24/40	LNA St3 PTAT bias control in Thru/Split Modes
d2a_CH<X>_<H/V>_CTAT	0	Quad CTAT bias control
d2a_DIST_B0_St1_cbias	0	Offset bias control
d2a_DIST_B0_St2_0_cbias	0	Offset bias control
d2a_DIST_B1_St1_cbias	0	Offset bias control
d2a_DIST_B1_St2_0_cbias	0	Offset bias control
d2a_DIST_CTAT	0	DIST CTAT bias control
d2a_DIST_B<X>_St1_PTAT	20	DIST St1 PTAT bias control
d2a_DIST_B<X>_St2_0_PTAT	20	DIST St2_0 PTAT bias control
d2a_DIST_B<X>_St2_1_PTAT	20	DIST St2_1 PTAT bias control
d2a_extra_bits_B0	0	Capacitance tune calibration for center combiners; <3:0> for the St2 combiners and <7:4> for the St1 combiners at Beam 0;
d2a_extra_bits_B1	0	Capacitance tune calibration for center combiners; <3:0> for the St2 combiners and <7:4> for the St1 combiners at Beam 1;
d2a_CH0_H_capture	0	Capacitance tune calibration for CH0 H; <3:0> for the St2 LNA and <7:4> for the St3 splitter;
d2a_CH0_V_capture	0	Capacitance tune calibration for CH0 V; <3:0> for the St2 LNA and <7:4> for the St3 splitter;
d2a_CH1_H_capture	0	Capacitance tune calibration for CH1 H; <3:0> for the St2 LNA and <7:4> for the St3 splitter;
d2a_CH1_V_capture	0	Capacitance tune calibration for CH1 V; <3:0> for the St2 LNA and <7:4> for the St3 splitter;
d2a_CH2_H_capture	0	Capacitance tune calibration for CH2 H; <3:0> for the St2 LNA and <7:4> for the St3 splitter;
d2a_CH2_V_capture	0	Capacitance tune calibration for CH2 V; <3:0> for the St2 LNA and <7:4> for the St3 splitter;
d2a_CH3_H_capture	0	Capacitance tune calibration for CH3 H; <3:0> for the St2 LNA and <7:4> for the St3 splitter;
d2a_CH3_V_capture	0	Capacitance tune calibration for CH3 V; <3:0> for the St2 LNA and <7:4> for the St3 splitter;

6 Device Information

6.1 Device Description

Table 11 shows the die size of the 5.2 mm x 5.2 mm CSP package.

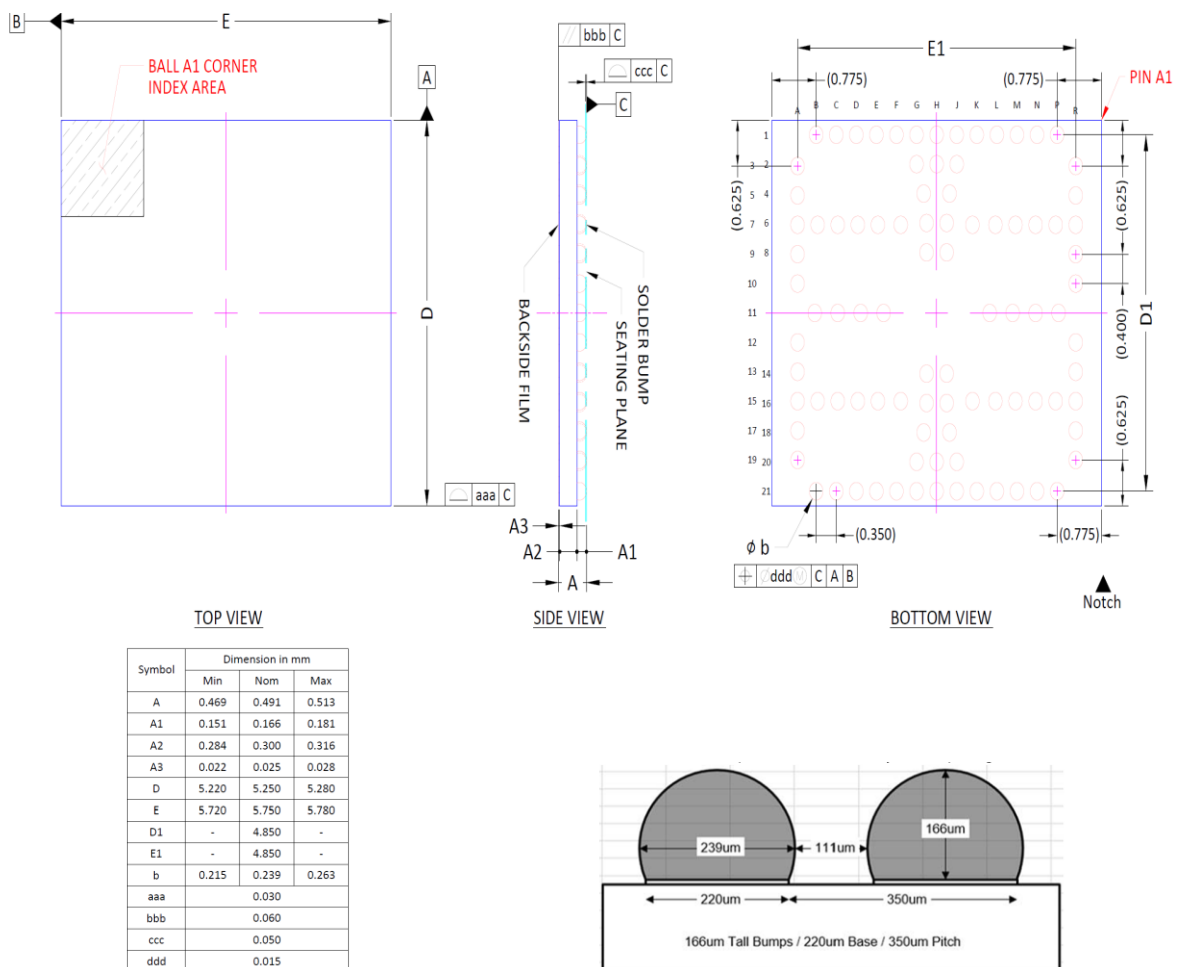
Table 11: Device Information

Part Number	Package	Body Size
	CSP	5.2 mm x 5.2 mm

6.2 Solder Bump Discussion with Suggested Reflow Profiles

Typical peak reflow temperature target range for SAC305 solder paste is 240 C to 250 C. SAC305 is one of the most commonly used paste materials in the industry.

Figure 6.2.1. The Ball pitch dimensions



6.3 Power Supply Sequencing

For Power-Up Sequencing:

- “VDD_DIG” (or “1V” if they are merged) should be raised to 1 V first.
- “VDD_1p8_IO” (or “1.8V” if they are merged) should be raised to 1.8V. At this point, the chip should respond to SPI commands.
- Following this, “VDD_1p8_ANA” supplies (if not merged) can be raised to 1.8V.
- Following this, the “VDD_FE1”, “VDD_FE3”, and “VDD_1p5V_DIST” supplies can be raised to 1V.
- Following this, the “VDD_FE3” and “VDD_1p5V_DIST” supplies can be raised to 1.5V from 1V.

For Power Down Sequencing,

1. “VDD_1p8_ANA” and “VDD_1p8_IO” rails can be transitioned from 1.8 to 1.5V.
2. The “VDD_FE3,” “VDD_1p5V_DIST,” “VDD_1p8_ANA,” and “VDD_1p8_IO” rails need to be transitioned from 1.5V to 1V.
3. All rails, including “VDD_FE1” and “VDD_DIG,” need to be transitioned from 1V to 0V.